Laila Assy

Oakville, ON | laila.assy@mail.utoronto.ca | linkedin.com/in/laila-assy | github.com/lailaassy

EDUCATION

University of Toronto 2023 – 2029 + PEY

B.A.Sc. in Computer Engineering + PEY Co-op, Dean's Merit Entrance Award *Toronto, ON*

EXPERIENCE

Developer Co-op, Visibility & Telemetry Team May 2026 – Present

Royal Bank of Canada *Toronto, ON*

- Cut incident investigation time from **∼3 hours to ∼1 hour** for **250+ network engineers** by building a Network Operations AI agent connected to **7 MCP servers**, including internal tools.
- Made network health clear to non-technical app owners in **under 1 minute** by creating **TCP Insights**, with AI summaries, dashboards, likely causes, and downloadable reports.
- Reduced agent response time from **∼9 minutes to under 1 minute** by auditing Langfuse traces for bottlenecks, while shipping releases at **80%+ test coverage** through SonarQube and GitHub Actions CI/CD.
- Reached **300+ users** across Slack, Webex, and Exchange by designing and launching a multi-tenant Kafka and webhook notification platform with tenant isolation, retries, and dead-letter queues.

Pro Bono Technical Consultant Oct 2026 – Present

Prospera Labs, via UofT Engineering Consulting Association *Toronto, ON*

- Building the ingestion and chunking layer of a **RAG backend for Shopify AI agents**, and testing retrieval quality so answers stay grounded in store product data.

Undergraduate Research Assistant Jul 2025 – Nov 2025

Behavior-Centric Cybersecurity Center, York University *Toronto, ON*

- Identified insider schemes as the leading fraud pattern on Solana by building and analyzing a dataset of **50+ incidents** and **100+ scam-linked addresses** by location and attack type.

IT Intern May 2025 – Aug 2025

Zarr Tech *Toronto, ON*

- Saved **$2,000+ per year** in maintenance costs by planning an on-premises to **Azure cloud migration**, keeping systems available so operations ran without interruption.
- Resolved **100+ support tickets** using Datto RMM, Kaseya Autotask, and TeamViewer, and deployed patches and updates to reduce security exposure.

PROJECTS

ClearRoute Accessible Navigation Map Jan 2026 – May 2026

University of Toronto | C++, OpenStreetMap, OpenMP

- Co-led a 3-person team building navigation for **low-vision users** following **WCAG** accessibility guidelines, with high-contrast themes, text-to-speech directions, fuzzy search, and multi-stop route planning.
- Achieved a **2x routing speedup** within a **50-second limit** by implementing A* search and parallelizing it with OpenMP across 8 cores.

Workflow Optimization Consulting Project Jan 2024 – Apr 2024

Toronto Seniors Services and Long-Term Care

- Projected a **66% efficiency gain** by mapping staffing and scheduling pain points with City of Toronto stakeholders over 4 months and presenting 3 budgeted recommendations.

LEADERSHIP & ACTIVITIES

Multimedia Director May 2026 – Present

IEEE Student Branch UofT *Toronto, ON*

- Grew the branch's audience by **200+ followers** and **10K+ views** by leading a 4-person team that produced **20+ posts** and promoted events that all sold out.

Marketing Director Aug 2024 – Jan 2026

UofT Engineering Kompetitions *Toronto, ON*

- Increased participation by **50%** to **300+ attendees** by leading marketing campaigns and sponsorship outreach.

SKILLS

Languages: Python, TypeScript, JavaScript, C/C++, SQL, Verilog, Assembly, HTML/CSS, R, MATLAB

AI/Software: LangGraph, LangChain, Langfuse, MCP Servers, RAG, Prompt Engineering, React, Flask, Node.js, REST APIs

Quality/Testing: Agent Behavior Analysis, Smoke Testing, SonarQube, CI/CD, GitHub Actions, Telemetry Analysis, Grafana, ExtraHop

Cloud/Infra: OpenShift, Azure, Git, Containerization, Kafka, Webhooks

Product/Tools: Agile/Scrum, Jira, Figma, User-Centered Design, Data Visualization, Stakeholder Collaboration